

CLAIM - Alpha as a 21|30 Closure Gate

First presentation: alpha as a closure margin from one named gate, not a fitted decimal.

This note claims that the fine-structure constant can be read as a closure margin produced by a specific 21|30 gate. The claim is conditional on a C30 carrier and a 21-step readout window; the non-trivial part is that this pair forces the rest of the ledger. Reading a 21-window on a C30 carrier forces a three-chain split, the rho=10 decagon cycle, the beat closure 210, and the first full active closure 1260.

structure

$$\alpha_{RT} = \frac{\pi}{30 \cdot 1260} (3 \cdot 30 - \Delta)$$

correction grammar

$$\Delta = AB + \frac{AB}{\rho} - \frac{1}{\rho G} - \frac{D}{\rho GCS}$$

expanded expression and bindings

$$\Delta = 2 + \frac{2}{10} - \frac{1}{10 \cdot 39} - \frac{20/21}{10 \cdot 39 \cdot 30 \cdot 7}$$
$$AB = 2, \quad \rho = 10, \quad G = 3(14 - 1) = 39, \quad D = \frac{20}{21}, \quad C = 30, \quad S = 7$$

numerical check / two-value status

$$\alpha_{RT} = 0.0072973525630485$$
$$\Delta(\alpha^{-1}) \approx 2.41 \times 10^{-8}$$

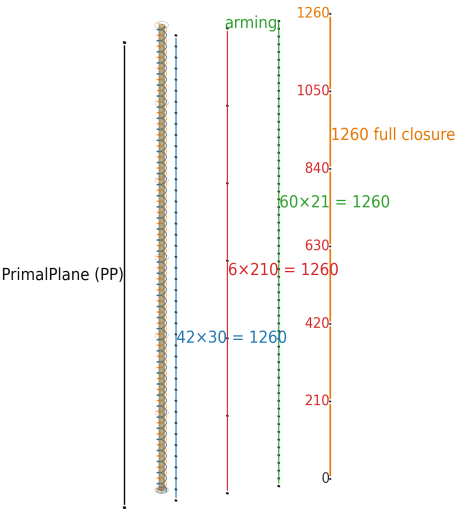
$$\alpha_{RT}^{-1} = 137.035999201092$$
$$\alpha_{CODATA\ 2022}^{-1} = 137.035999177(21)$$

Gate value = ideal closure margin; CODATA = operational laboratory value. Offset = explicit target, not hidden rounding.

Full closure

$$1260 = 21 \times 3 \times 20 = 6 \times 210 = 42 \times 30$$

Full closure rule: first N where 3 chains carry 20 active windows each.
Thus $N/21 = 3 \times 20 = 60$, so $N = 1260$.

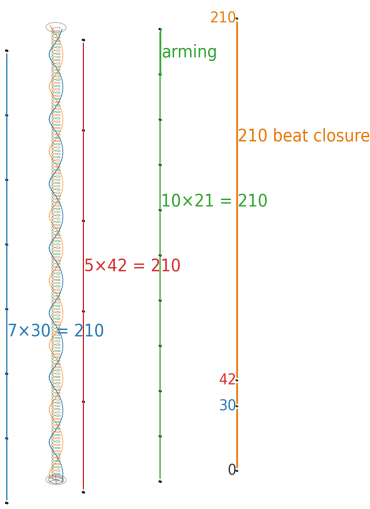


Bars and gradation ticks retained: 42 x 30, 6 x 210, 60 x 21.

Beat closure

$$210 = 7 \times 30 = 5 \times 42 = 10 \times 21$$

210 is the local beat: $\text{lcm}(21,30)=210$.
10 windows: $210/21=10$; 7 C30 legs close with 5 forty-two waves.



Local cell where the 21|30 gate becomes visible.

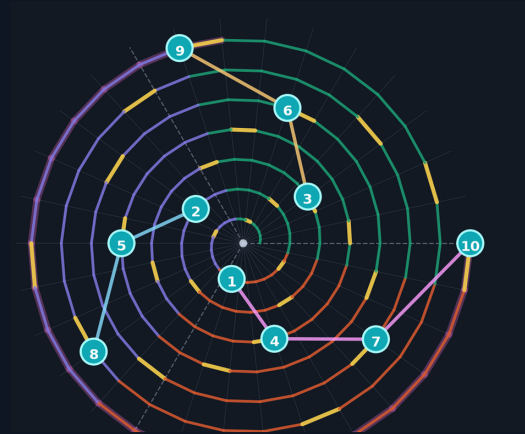
Break condition Identify a link invalid, non-unique under stated rules, or replaceable by a stricter construction.

THE 21|30 GATE - Break the Ledger

The point is not that the numbers are interesting; the point is that the links co-lock.

One beat: the 21|30 gate

Intentionally non-physical display. Geometry is unfolded for readability.
Visual aid only; the claim stands or falls on the ledger links.



spiral arms retained: $210 = 10 \times 21 = \text{lcm}(21,30)$

W21 on C30

read a 21-window on a 30-slot carrier

$\text{gcd}(21,30) = 3$

forced split into three transport chains

$30 / 3 = 10$

rho / decagon readout, 36 degrees per chain step

$\text{lcm}(21,30) = 210$

one local beat closure

$N = 21 \times 3 \times 20 = 1260$

full closure rule: first N with 3 chains x 20 active positions

$G = 3(14 - 1) = 39$

$42 = 3 \times 14$, then one seam excluded per chain

Companion visualizer (not part of the claim): package file `RT_ALPHA_READOUT_GEOMETRY.html`; GitHub path: https://github.com/Bo-Fremling/rhythm-theory-release/blob/main/RT_ALPHA_READOUT_GEOMETRY.html. It is only an interactive reader aid; the claim stands or falls on the ledger links below.

Gate link	Consequence
$3 = \text{gcd}(21,30)$	forced split into three transport chains
$10 = 30 / 3$	rho = 10 / decagon readout
$210 = \text{lcm}(21,30)$	one local beat closure
$1260 = 21 \times 3 \times 20 = 6 \times 210 = 30 \times 42$	full closure rule: first N with 3 chains x 20 active positions
$42 = 1260 / 30 = 3 \times 14$	C30 carrier turns under the three-chain split
$G = 3 \times (14 - 1) = 39$	one excluded seam per transport chain

Correction grammar, not term fitting

Generated in order: A/B polarity -> rho-readout -> seam-gap subtraction -> final gate-intersection.

$\Delta = AB + AB/\rho - 1/(\rho G) - D/(\rho G C S)$

Allowed factors must already be forced by the 21|30 gate: $AB=2$, $\rho=10$, $G=39$, $D=20/21$, $C=30$, $S=7$.

Expansion: $\Delta = 2 + 2/10 - 1/(10 \times 39) - (20/21)/(10 \times 39 \times 30 \times 7)$.

Critical link: $42=3 \times 14$; one seam excluded per chain gives $G=3 \times (14-1)=39$.

A stricter competing grammar would refute or improve this claim.

Final claim

alpha_RT is proposed as a closure-geometric entity, structurally analogous to pi in the limited sense that both are dimensionless closure measures: pi measures circular closure; alpha_RT measures closure defect in the RT release ledger. To refute it, break one named link under the stated rules. Non-uniqueness test: a different primitive gate must generate the same closure value using only its own forced links and an equal-or-stricter grammar; free recombinations do not count.